Planck-to-Cosmos Observation Rulers

Typed logarithmic charts, scale passports, and cosmological landmarks

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Status. This document supersedes *Planck-to-Cosmos Observation Rulers v1.0.* It is a reference module for GO Core v0.2 and the Distance-Scale Interface v0.2. The revision separates intrinsic system scales from instrument resolutions, treats Planck units as normalizations rather than cutoffs, and treats cosmic entries as model-, epoch-, and convention-dependent landmarks. The logarithmic chart is standard mathematics; no new fundamental law is claimed.

Abstract

For a positive length, invariant mass, and duration, a declared reference triple $\boldsymbol{a} = (a_L, a_M, a_T)$ and base $b > 1$ define the logarithmic chart

$$\boldsymbol{\chi}_{b,\boldsymbol{a}}(L, M, T) = \left(\log_b \frac{L}{a_L}, \log_b \frac{M}{a_M}, \log_b \frac{T}{a_T}\right).$$

This map is a global smooth chart from the positive scale cone $(\mathbb{R}_{>0})^3$ to $\mathbb{R}^3$. Coherent unit changes leave it invariant, anchor changes translate it, and base changes act by a scalar linear map. Coordinate differences support a family of weighted log-scale metrics; there is no canonical physical reason to assign equal weight to one decade of length, mass, and time. Intrinsic scale catalogues belong to the system descriptor ξ, whereas spatial, mass-equivalent, and temporal resolutions belong to an observation protocol λ. Their anchor-independent comparison is the resolution margin $g_i = \log_b(q_i/\varepsilon_i)$. Monomial physical relations become linear constraint surfaces in log coordinates. In Planck normalization, the causal condition $L \le cT$ gives $\chi_L \le \chi_T$, the reduced Compton relation gives $\chi_L = -\chi_M$, and the Schwarzschild relation gives $\chi_L = \chi_M + \log_b 2$. A reproducible cosmological ledger then shows why a bare "mass of the observable Universe" is not a typed quantity: baryonic, total-matter, and total-energy mass equivalents differ by more than a decade under the same rounded horizon convention.

Contents

1 Scope and claim boundary

The original ruler note correctly normalized physical quantities before taking logarithms. Its unresolved issue was semantic rather than algebraic: object sizes, instrument resolutions, and cosmological landmarks were placed in the same coordinates without distinct passports. The notation

$$Y_X(s, \mu, \tau) = O_{s,\mu,\tau}(X)$$

also treated every point of $\mathbb{R}^3$ as if it automatically specified an observation channel. It does not. A channel requires a declared measurement model, frame, calibration, noise law, estimator, and applicable axes.

Boundary 1.1 (Claim firewall). The Planck length and Planck time are not asserted to be experimentally proved minimum scales. The present comoving particle horizon is not the diameter of the entire Universe. Similar numerical spans of roughly sixty decades do not imply a new physical law. A camera, clock, or spectrometer need not measure all of length, invariant mass, and time.

The strict claim is limited to the following:

> Positive physical scales admit useful typed logarithmic coordinates. Systems generally determine labelled sets or regions of such coordinates, while observation protocols determine separate resolution coordinates and response traces.

2 Typed scale data and the two-passport rule

Definition 2.1 (Scale item)**.** A scale item is a record

$$\sigma = (\mathsf{id}, \mathsf{axis}, q, \mathsf{role}, \mathsf{frame}, \mathsf{convention}, \mathsf{U}),$$

where $\mathsf{axis} \in \{L, M, T\}$, $q > 0$ has the corresponding physical dimension, role identifies the measurand, and U contains exactness or uncertainty information.

The role is indispensable. A particle radius, a wavelength, a correlation length, and a detector pixel pitch all have dimension L, but they are not interchangeable. Likewise, invariant rest mass is not relativistic energy divided by an undeclared constant, and proper time is not a coordinate duration without a clock congruence.

Definition 2.2 (Intrinsic scale catalogue)**.** For a system X, let

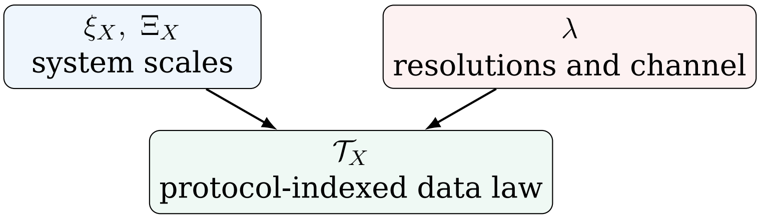
$$\Xi_X = \{\sigma_\alpha : \alpha \in A_X\}$$

be a labelled, possibly partial and multivalued catalogue of intrinsic or model-derived scales. It is a component of the system descriptor ξ_X. A missing axis is recorded as `not-applicable`; it is not encoded by 0 or $+\infty$.

Definition 2.3 (Observation protocol passport)**.** An observation protocol is a tuple

$$\lambda = \big(R_\lambda, O_\lambda, K_\lambda, Q_\lambda, \boldsymbol{\varepsilon}_\lambda, \boldsymbol{r}^*_\lambda, \mathsf{F}_\lambda, \widehat{\theta}_\lambda, \mathsf{U}_\lambda\big).$$

Here R_λ is a reduction, O_λ is an ideal deterministic observation, K_λ is a stochastic kernel or an explicit noiseless flag, Q_λ is a quantizer or partition, $\boldsymbol{\varepsilon}_\lambda$ contains only applicable resolutions, $\boldsymbol{r}^*_\lambda$ contains their reference scales, F_λ records frames and conventions, $\widehat{\theta}_\lambda$ is an estimator or declared direct readout, and U_λ records uncertainty.



Remark 2.4. A mass axis in λ exists only if the channel measures mass, invariant mass, or a declared mass-equivalent. If the actual threshold is an energy ε_E and the model uses $E = Mc^2$, then

$$\varepsilon_M = \frac{\varepsilon_E}{c^2}$$

is a typed converter. Without such a converter, "mass resolution" is not available.

3 The logarithmic scale chart

Definition 3.1 (Positive scale cone and chart)**.** Let

$$\mathcal{S}_{LMT} := \mathbb{R}^{(L)}_{>0} \times \mathbb{R}^{(M)}_{>0} \times \mathbb{R}^{(T)}_{>0}.$$

For a positive reference triple $\boldsymbol{a} = (a_L, a_M, a_T)$ and $b > 1$, define

$$\boldsymbol{\chi}_{b,\boldsymbol{a}}(L, M, T) = \begin{pmatrix} \chi_L \\ \chi_M \\ \chi_T \end{pmatrix} := \begin{pmatrix} \log_b(L/a_L) \\ \log_b(M/a_M) \\ \log_b(T/a_T) \end{pmatrix}.$$

Theorem 3.2 (Global chart)**.** *The map $\boldsymbol{\chi}_{b,\boldsymbol{a}} : \mathcal{S}_{LMT} \to \mathbb{R}^3$ is a smooth diffeomorphism with inverse*

$$\boldsymbol{\chi}^{-1}_{b,\boldsymbol{a}}(x_L, x_M, x_T) = \big(a_L b^{x_L}, a_M b^{x_M}, a_T b^{x_T}\big).$$

Proof. Each one-dimensional map $q \mapsto \log_b(q/a)$ is smooth, strictly increasing, and bijective from $\mathbb{R}_{>0}$ to $\mathbb{R}$, with the displayed smooth inverse. The product map has the same properties. □

The chart is an exact reparameterization, not a reduction. Information loss enters when a scale item is selected from a system, rounded, thresholded, or passed through an observation channel.

Definition 3.3 (Scale region)**.** For a catalogue Ξ_X, a declared extraction rule E_{r} selects compatible triples or partial tuples. Its logarithmic image is

$$\mathscr{R}_{X,\mathsf{r}} = \boldsymbol{\chi}_{b,\boldsymbol{a}}\big(E_{\mathsf{r}}(\Xi_X)\big).$$

An extended or multiscale object is represented by this labelled set or region, not by a forced single point.

Boundary 3.4 (No canonical triple). An atom has several length scales and several characteristic times. A galaxy has luminous, half-mass, virial, and halo radii. Selecting one value on each axis is a protocol or modelling decision. The phrase "the object occupies the point (χ_L, χ_M, χ_T)" is valid only after the extraction rule has been declared.

4 Unit, anchor, and base transformation laws

Theorem 4.1 (Coordinate covariance)**.** *Let $q, a > 0$ have the same physical dimension.*

(i) Under a coherent unit conversion $(q, a) \mapsto (\alpha q, \alpha a)$ with $\alpha > 0$, $\log_b(q/a)$ is unchanged.

(ii) If the reference changes from a to $a' > 0$, then

$$\chi_{b,a'}(q) = \chi_{b,a}(q) - \log_b \frac{a'}{a}.$$

(iii) If the base changes from b to $b' > 1$, then

$$\chi_{b',a}(q) = \frac{\ln b}{\ln b'}\, \chi_{b,a}(q).$$

Proof. The first statement follows from cancellation of α. The second is the logarithm of $(q/a)/(a'/a)$. The third follows from the change-of-base formula. □

Thus an anchor change translates log-space. Coordinate differences, axis orderings, and any construction depending only on differences are anchor independent. A change from decades to octaves or nats rescales the chart.

Definition 4.2 (Weighted log-scale distance)**.** For a symmetric positive-definite dimensionless matrix $W \in \mathbb{R}^{3\times 3}$, define

$$d_{W,b}(\boldsymbol{q}, \boldsymbol{q}') := \sqrt{\big(\boldsymbol{\chi}_{b,\boldsymbol{a}}(\boldsymbol{q}) - \boldsymbol{\chi}_{b,\boldsymbol{a}}(\boldsymbol{q}')\big)^{\mathsf{T}} W \big(\boldsymbol{\chi}_{b,\boldsymbol{a}}(\boldsymbol{q}) - \boldsymbol{\chi}_{b,\boldsymbol{a}}(\boldsymbol{q}')\big)}.$$

Theorem 4.3 (Metric and transformation law)**.** *The function $d_{W,b}$ is a metric on $\mathcal{S}_{LMT}$, independent of the anchor $\boldsymbol{a}$ and invariant under coherent unit changes. Under a base change $b \mapsto b'$, it remains numerically invariant if*

$$W_{b'} = \left(\frac{\ln b'}{\ln b}\right)^2 W_b.$$

Proof. The chart in theorem 3.2 is injective, and the positive-definite quadratic form defines a norm on coordinate differences. Anchor translations cancel in those differences. Unit invariance follows from theorem 4.1. If $c_{bb'} = \ln b/\ln b'$, then $\Delta\boldsymbol{\chi}_{b'} = c_{bb'}\Delta\boldsymbol{\chi}_b$; the displayed transformation of W cancels $c_{bb'}^2$. □

Boundary 4.4 (Metric convention). $W = I$ declares that one decade of length, mass, and time contributes equally. This may be useful for visualization, but it is not a physical law. A scientifically interpretable W must come from a decision problem, tolerances, or a covariance model.

5 Descriptor coordinates versus resolution coordinates

The Distance-Scale Interface uses a resolution coordinate whose orientation is opposite to the descriptor coordinate.

Definition 5.1 (Protocol resolution depth)**.** For an applicable protocol axis i with resolution $\varepsilon_i > 0$ and reference $r_i^* > 0$, define

$$\eta_i(\lambda) := \log_b \frac{r_i^*}{\varepsilon_i}.$$

Finer resolution increases η_i. If $r_i^* = a_i$, then

$$\eta_i = -\chi_{b,a_i}(\varepsilon_i).$$

Definition 5.2 (Resolution margin)**.** For a system scale $q_i > 0$ and protocol resolution $\varepsilon_i > 0$ on the same typed axis, define

$$g_i(q_i, \lambda) := \log_b \frac{q_i}{\varepsilon_i}.$$

Proposition 5.3 (Anchor-independent comparison)**.** *For any common positive anchor a_i,*

$$g_i = \chi_{b,a_i}(q_i) - \chi_{b,a_i}(\varepsilon_i).$$

Hence g_i is invariant under coherent unit changes and common anchor changes.

Proof. Subtract the two logarithms and combine their arguments. □

In a simple scale-separation model, $g_i > 0$ means that the intrinsic scale exceeds the nominal resolution. It does not by itself specify contrast, signal-to-noise ratio, inverse-problem conditioning, or a detection probability.

Definition 5.4 (Protocol-indexed observation trace)**.** For a declared family Λ of protocols, a deterministic trace is

$$\mathcal{T}_X^{\text{det}} = \left\{ \left(\boldsymbol{\eta}(\lambda), O_\lambda(X)\right) : \lambda \in \Lambda \right\}.$$

For a stochastic channel it is

$$\mathcal{T}_X^{\text{stoch}} = \left\{ \left(\boldsymbol{\eta}(\lambda), \text{Law}(Y_\lambda \mid X, \lambda)\right) : \lambda \in \Lambda \right\}.$$

The shorthand $O_{\boldsymbol{\eta}}$ is admissible only after the map $\boldsymbol{\eta} \mapsto \lambda(\boldsymbol{\eta})$ and all nuisance parameters have been defined.

6 Monomial algebra and constraint surfaces

Theorem 6.1 (Log-linearization of positive monomial laws)**.** *Let positive inputs $\boldsymbol{q} = (q_1, \ldots, q_n)$ and positive outputs r_i satisfy*

$$r_i = C_i \prod_{j=1}^{n} q_j^{A_{ij}}, \qquad r_i^* = C_i \prod_{j=1}^{n} (q_j^*)^{A_{ij}},$$

where the references are dimensionally matched. Then

$$\log_b \frac{r_i}{r_i^*} = \sum_{j=1}^{n} A_{ij} \log_b \frac{q_j}{q_j^*}.$$

In vector form, $\boldsymbol{\chi}_r = A\boldsymbol{\chi}_q$.

Proof. Divide the two monomials, cancel C_i, and apply the logarithm product and power laws. □

This theorem is the precise structural reason that dimensional monomials become hyperplanes in logarithmic coordinates. It is a coordinate simplification, not a claim that arbitrary nonlinear physics becomes linear.

6.1 Planck identities

Use the standard definitions

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad m_P = \sqrt{\frac{\hbar c}{G}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}.$$

Direct substitution gives

$$t_P = \frac{\ell_P}{c}, \qquad \ell_P m_P = \frac{\hbar}{c}, \qquad m_P c^2 = \frac{\hbar}{t_P}.$$

The three Planck anchors are therefore built from common constants and are not statistically or conceptually independent axes.

6.2 Causal, quantum, and gravitational examples

Corollary 6.2 (Causal half-space)**.** *If* L *and* T *refer to the same causal process and* $L \le cT$*, then in Planck normalization*

$$\chi_L \le \chi_T.$$

Proof. Because $\ell_P = ct_P$,

$$\frac{L/\ell_P}{T/t_P} = \frac{L}{cT} \le 1.$$

Apply the increasing logarithm. □

If L and T belong to unrelated catalogue entries, the hypothesis of corollary 6.2 fails and no such inequality follows.

Corollary 6.3 (Mass-energy, density, Compton, and Schwarzschild planes)**.** *In Planck normalization:*

$$\begin{aligned}
E = Mc^2, \quad E_P = m_P c^2 &\Longrightarrow \chi_E = \chi_M, \\
\rho = M/L^3, \quad \rho_P = m_P/\ell_P^3 &\Longrightarrow \chi_\rho = \chi_M - 3\chi_L, \\
\bar{\lambda}_C = \hbar/(Mc) &\Longrightarrow \chi_{\bar{\lambda}_C} = -\chi_M, \\
r_s = 2GM/c^2 &\Longrightarrow \chi_{r_s} = \chi_M + \log_b 2.
\end{aligned}$$

Proof. The first two identities follow from theorem 6.1. For the reduced Compton length, use $\ell_P m_P = \hbar/c$:

$$\frac{\bar{\lambda}_C}{\ell_P} = \frac{m_P}{M}.$$

For the Schwarzschild radius, $Gm_P/c^2 = \ell_P$, hence

$$\frac{r_s}{\ell_P} = 2\frac{M}{m_P}.$$

Taking logarithms proves the result. □

Boundary 6.4 (Physical-domain boundary). The lines in corollary 6.3 are relations among specifically coupled measurands. They do not turn the whole scale catalogue into one three-dimensional state space, and they do not by themselves establish a quantum-gravity model.

7 Planck anchors and coordinate uncertainty

Anchor	Definition	2022 CODATA value	Status
Planck length ℓ_P	$\sqrt{\hbar G/c^3}$	1.61625×10^{-35} m	normalization, not a proved minimum length
Planck mass m_P	$\sqrt{\hbar c/G}$	2.17643×10^{-8} kg	normalization, not a minimum mass
Planck time t_P	$\sqrt{\hbar G/c^5}$	5.39125×10^{-44} s	normalization, not a proved minimum duration

The 2022 CODATA values inherit uncertainty primarily through G; c and h are exact defining constants in the revised SI. Consequently, the Planck anchors are correlated. Treating their listed standard uncertainties as independent can give an incorrect covariance for derived scale coordinates.

Proposition 7.1 (First-order uncertainty of a log ratio)**.** *Let*

$$\chi = \log_b(q/a)$$

with $q, a > 0$*. To first order,*

$$u^2(\chi) = \frac{1}{(\ln b)^2}\left[\frac{u^2(q)}{q^2} + \frac{u^2(a)}{a^2} - 2\frac{\mathrm{Cov}(q,a)}{qa}\right].$$

For several correlated inputs,

$$\Sigma_\chi = J\Sigma_{q,a}J^\mathsf{T},$$

where J is the Jacobian of the logarithmic map.

Proof. Differentiate $\chi = (\ln q - \ln a)/\ln b$ and apply the first-order covariance propagation formula. □

The formula is a local approximation. Large, asymmetric, or model-posterior uncertainties should be propagated by the full distribution rather than by a symmetric linearized error bar.

8 Cosmological landmarks are typed model outputs

The phrase "observable-universe diameter" must name a cosmological distance. The legacy 93 Gly entry is retained only as a rounded *present comoving particle-horizon diameter* in a declared $a_0 = 1$ convention. It is not a light-travel distance and not the size of the entire Universe. Cosmological distance definitions are distinct even when they share the SI dimension of length.

The mass axis has a larger ambiguity. For a flat-volume illustrative ledger, let

$$\rho_{c0} = \frac{3H_0^2}{8\pi G}, \qquad V_{\text{flat}} = \frac{4\pi}{3} R_{\text{ph},0}^3.$$

Then three different mass-equivalent landmarks are

$$M_b = \Omega_b \rho_{c0} V_{\text{flat}}, \qquad M_m = \Omega_m \rho_{c0} V_{\text{flat}}, \qquad M_E = \rho_{c0} V_{\text{flat}}.$$

M_b counts the baryonic component, M_m all pressureless matter in the adopted parameterization, and M_E the total critical mass-equivalent. Dark energy is not a collection of rest-mass particles; the last quantity is explicitly an energy-equivalent bookkeeping scale.

Landmark	Value in SI	Coordinate	Convention
Present comoving particle-horizon diameter	8.79848×10^{26} m	61.7359	rounded 93 Gly landmark
FLRW cosmic age	4.35400×10^{17} s	60.9072	Planck 2018 baseline, 13.797 ± 0.023 Gyr
Baryonic mass-equivalent	1.50025×10^{53} kg	60.8384	$\Omega_b \rho_{c0} V$
All-matter mass-equivalent	9.58576×10^{53} kg	61.6439	$\Omega_m \rho_{c0} V$
Total-energy mass-equivalent	3.04310×10^{54} kg	62.1456	$\rho_{c0} V$

The table uses a rounded 93 Gly diameter and a Planck 2018 flat base-ΛCDM parameter ledger: $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_b = 0.0493$. Because the rounded horizon entry and the full parameter covariance are not supplied, no combined confidence interval is fabricated. The values are reproducible landmarks, not a precision cosmological analysis.

Proposition 8.1 (The legacy mass anchor is ambiguous)**.** *Under this single illustrative ledger,*

$$M_b \approx 1.50 \times 10^{53}\,\mathrm{kg}, \qquad M_m \approx 9.59 \times 10^{53}\,\mathrm{kg}, \qquad M_E \approx 3.04 \times 10^{54}\,\mathrm{kg}.$$

Their Planck-mass coordinates are approximately 60.84*,* 61.64*, and* 62.15*, respectively. Therefore an unqualified* 10^{53} kg *entry cannot serve as a unique cosmic mass anchor.*

Proof. Substitute the declared H_0, density fractions, and flat-volume convention into the preceding definitions, then divide by m_P and take $\log_{10}$. □

Boundary 8.2 (Epoch and model dependence). Changing the cosmological model, horizon definition, spatial curvature, epoch, or matter component changes these landmarks. They are catalogue records, not constants defining the logarithmic coordinate system.

9 Selected reproducible catalogue

The following table is generated from the machine-readable input ledger. Entries marked "order of magnitude" or "declared reference" are visual landmarks rather than measurements. Every coordinate uses the corresponding 2022 CODATA Planck anchor and base 10.

Entry	Axis	SI value	Coordinate	Status
planck length	length	1.6163×10^{-35} m	0.000	CODATA recommended
atomic order	length	1.0000×10^{-10} m	24.791	order of magnitude marker
human length reference	length	1.0000×10^{0} m	34.791	declared not population estimate
earth diameter	length	1.2742×10^{7} m	41.897	rounded mean diameter
astronomical unit	length	1.4960×10^{11} m	45.966	exact IAU definition
observable particle horizon diameter	length	8.7985×10^{26} m	61.736	rounded model dependent
electron mass	mass	9.1094×10^{-31} kg	-22.378	CODATA recommended
proton mass	mass	1.6726×10^{-27} kg	-19.114	CODATA recommended
planck mass	mass	2.1764×10^{-8} kg	0.000	CODATA recommended
human mass reference	mass	7.0000×10^{1} kg	9.507	declared not population estimate
earth mass	mass	5.9722×10^{24} kg	32.438	rounded astronomical value
solar mass	mass	1.9885×10^{30} kg	37.961	rounded astronomical value
planck time	time	5.3912×10^{-44} s	0.000	CODATA recommended
femtosecond	time	1.0000×10^{-15} s	28.268	SI prefix exact
second	time	1.0000×10^{0} s	43.268	SI base unit
day	time	8.6400×10^{4} s	48.205	conventional exact duration
Julian year	time	3.1558×10^{7} s	50.767	conventional exact duration
universe age	time	4.3540×10^{17} s	60.907	model inferred parameter

The equality of the light-year length coordinate and the Julian-year time coordinate to the displayed precision is not mysterious: $1\,\mathrm{ly} = c \times 1\,$Julian year and $\ell_P = ct_P$. It is an exact consequence of matched definitions.

10 Normative reporting contract

For a static catalogue, stochastic-channel, quantizer, temporal resolution, and observation-horizon fields may be `not-applicable`; the field must still be present. For an actual measurement trace, applicable fields require values rather than prose placeholders.

Field	Type	Required declaration
Hidden system X	state space or object	domain and admissible states
System descriptor ξ_X	typed record	intrinsic model data and scale catalogue Ξ_X
Scale item	typed positive quantity	axis, role, frame, convention, value or interval, uncertainty

Field	Type	Required declaration
Extraction rule	reduction	which labelled items form a tuple or region
Reference triple $\boldsymbol{a}$	positive typed tuple	Planck or alternative anchors, with source
Logarithm base b	dimensionless, $b > 1$	decades, octaves, or other declared base
Observation protocol λ	channel passport	reduction, observation, noise/exactness, quantizer, estimator
Applicable axes	subset of $\{L, M, T\}$	missing axes recorded as not applicable
Spatial resolution	positive length or not applicable	frame and calibration
Mass resolution	positive mass or not applicable	direct channel or explicit energy/mass converter
Temporal resolution	positive duration or not applicable	clock and frame convention
Observation horizon	duration or not applicable	acquisition interval
Reference scales $\boldsymbol{r}^*$	positive typed tuple	normalization of resolution depth
Resolution margin	dimensionless	matched measurand and resolution
Frame passport	metadata	proper, coordinate, rest, comoving, or other declared convention
Cosmology passport	model metadata or not applicable	model, epoch, distance, volume, and component definitions
Uncertainty	covariance, law, or exact flag	dependence and propagation semantics
Weight matrix W	positive definite or not applicable	rationale and base-transformation policy
Defect normalization	typed ratio or not applicable	required before scalar loss aggregation

11 Claim audit

Statement	Status	Necessary hypotheses
$\boldsymbol{\chi}_{b,\boldsymbol{a}}$ is a global chart on $(\mathbb{R}_{>0})^3$	theorem	positive scales, positive anchors, $b > 1$
Log coordinates are invariant under a unit change	theorem	quantity and anchor converted coherently
Anchor changes preserve coordinate differences	theorem	common typed anchor change
$d_{W,b}$ is a scale-space metric	theorem	W symmetric positive definite
$W = I$ is the unique physical metric	false	no such canonical choice is proved
Every system has one canonical (L, M, T) point	false	requires a declared extraction rule

Statement	Status	Necessary hypotheses
Every protocol measures all three axes	false	missing axes are allowed
$g_i > 0$ guarantees detection	false	only a simple scale-separation diagnostic
Planck units are minimum physical scales	not claimed	they are used as normalizations
The causal inequality gives $\chi_L \le \chi_T$	corollary	paired length and time of the same causal process
Compton and Schwarzschild relations become affine lines	corollary	declared standard formulas and Planck normalization
93 Gly is the diameter of the entire Universe	false	rounded present comoving particle-horizon convention only
10^{53} kg is the unique mass of the observable Universe	false	component, volume, epoch, and model are required
Three spans near sixty decades imply a new law	not claimed	numerical similarity depends on selected landmarks

12 Conclusion

The strict Planck-to-cosmos ruler object is not a bare point in $\mathbb{R}^3$. It is the pair

$$\big(\xi_X, \Xi_X; \lambda, \boldsymbol{\varepsilon}_\lambda, \boldsymbol{r}^*_\lambda\big),$$

together with a positive reference tuple, logarithm base, frame and cosmology conventions, uncertainty model, and a protocol-indexed observation trace. The descriptor chart

$$\boldsymbol{\chi}_{b,\boldsymbol{a}}(\boldsymbol{q}) = \log_b(\boldsymbol{q}/\boldsymbol{a})$$

is unit invariant and turns positive monomial relations into linear constraints. The resolution margin $\log_b(q_i/\varepsilon_i)$ compares a typed system scale with a typed protocol resolution without conflating the two. Cosmic quantities are optional model outputs. This separation closes the central blocker in v1.0 and makes the module compatible with the GO Core distance-scale interface.

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